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Plethora of Carbon Nanotubes Applications in Various Fields – A State-of-the-Art-Review

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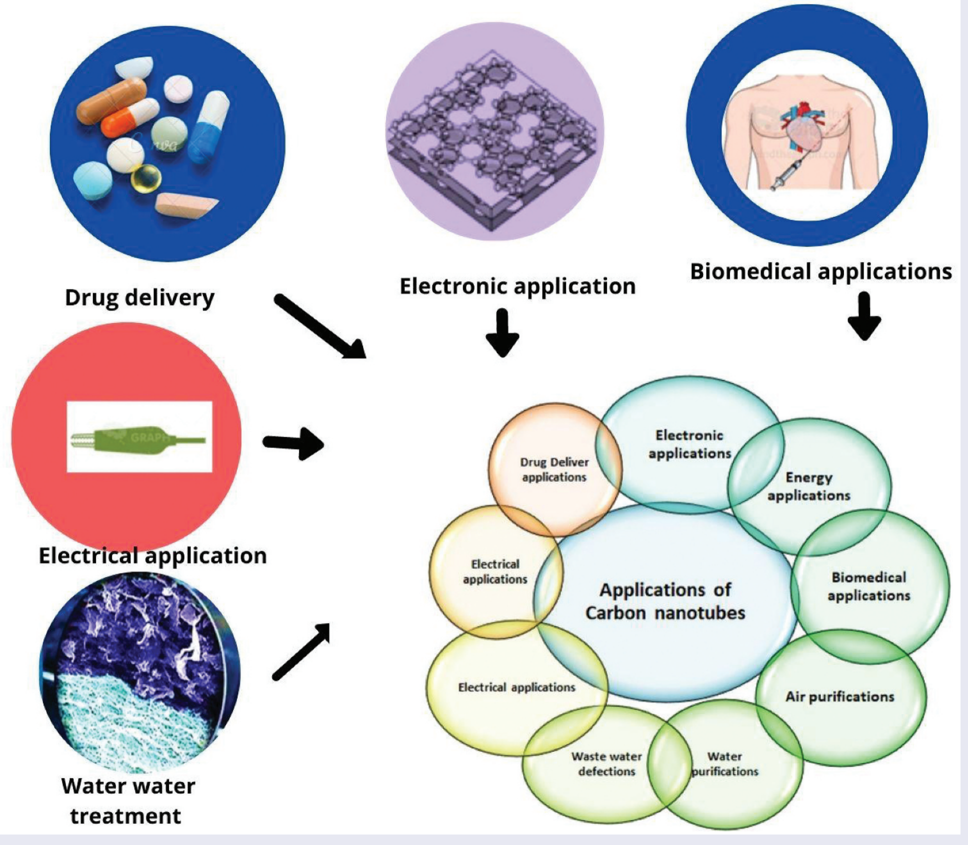
ABSTRACT
Carbon nanotubes diverse applications have been discussed in the present study. Carbon nanotubes are a promising material in all façades of life, be it environment, electrical, electronics, biomedical energy, food technology applications, etc. Carbon nanotubes open new avenues to improve the environment, control pollution, and used in various fields. Increase in the population, urbanization and industrialization create several types of pollution. Carbon nanotubes solve various problems including environmental issues, pollution, bio-medicals, etc. It becomes an inevitable material for the present and future. In this review article, carbon nanotube applications to solve water purification, water disinfection, electrical and electronic applications, biomedical energy, or in-air purification have been discussed. A plethora of carbon nanotube applications in various fields have been discussed in articles covering the fields of water purification, biomedical applications in electrical/electronic applications, agricultural yields, etc. Eleven different case studies have been discussed in the paper and testimony of the use of CNTs in various fields was given by the patent trend.



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1. Introduction

Carbon nanotubes (CNTs) are constantly being used for environment protection. Single walled (SWCNTs) and multiwalled carbon nanotubes (MWCNTs) were used for the preponderance of nanoscale carbon production. A single sheet of graphene is used to synthesize single-walled carbon nanotubes, which have a cylindrical design. Under a transmission electron microscope, it appeared as two planes. Carbon nanotubes with several walls consisting of multiple layers of graphene arranged in a concentric shape all around the smallest nanotube. Pure water and air is the utmost important requirement for the survival of human beings. Water is being polluted day by day increase and over utilization of the resources. Air, which we breathe and exhaled, is continuously polluted because of the excess use of nonrenewable resources such as petrol and diesel, etc. Increase of demand of energy leads to industrialization and alternative resources. CNTs have astonishing physical, chemical, and electronic properties which act as cost effective in various applications. Reduction of waste disposal in the environment can be achieved by using carbon nanotubes as super capacitors which provide small size, high portability, maximum power density, long suitability, and high energy [1–5]. The hybrid materials manufactured by carbon nanotubes and carbon nanofibers in blend with inorganic (metal oxide) nanoparticles can be able to solve the maximum number of water treatment problems, helpful in preventing air pollution, and recycling of waste [6–10].

CNT sponges have high porosity and very low density and have haphazardly entangled three-dimensionally, which is used for removing oil from water [11–14]. Camphor is known as an environment-friendly hydrocarbon. High-purity carbon nanotubes (CNTs) can be produced from camphor hydrocarbons using the chemical vapor deposition method. The synthesis technique depicted here fully addresses the '12-principle' protocol of green chemistry [15–19].

Energy generation, which generally occurs through fossil fuels, has been recognized as the major issue for environmental degradation from global warming. Energy storage is also a hot issue for rapid growth in developing nations like India. Bio-medically, there is always a substantial demand of alternative materials for drug delivery, cancer treatment, biosensors, etc. To address these issues, carbon nanotubes act as a revolutionizing material for energy conversion and storage materials. And, it helps in reducing environment pollution in air and water media because of their exclusive structural, electronic, and mechanical properties [20–24].

Here, the authors can review the applications of carbon nanotubes to solve water purification, water disinfection, electrical and electronic applications,

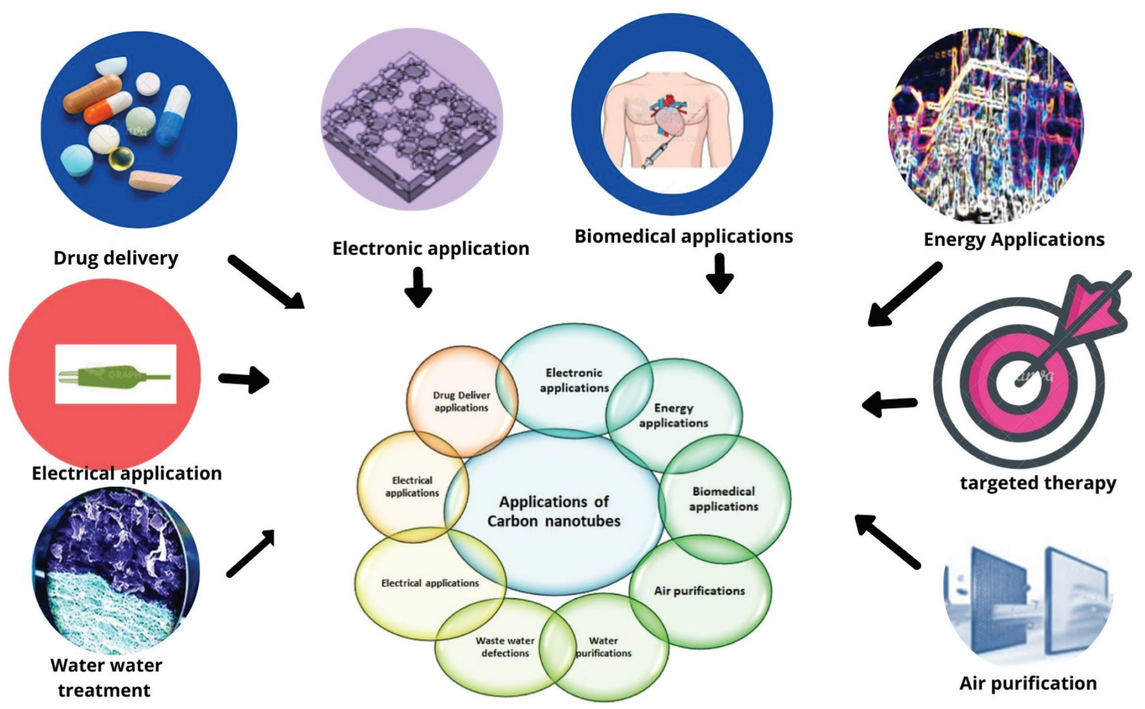

Figure 1. The various environmental applications of carbon nanotubes have been depicted in the figure, which includes water purification, waste water disinfection, electrical, electronic, energy, biomedical, and air purification.

future smart electronics, sensor tapes, and capacitors, and batteries, biomedical, energy, or in-air purification, etc. Figure 1 represents the various environmental applications of carbon nanotubes. Technology and equipment run by electricity is a vital issue in our daily life. Among a range of electrochemical energy storage Li–S batteries are a capable contender for quenching the thrust of energy for the next generation [25–35].

CNTs are used for delivering bio-fertilizers. When bio-fertilizer is delivered with the help of CNTs, the unique properties of its provided large surface area, superior reactivity, and smaller particle size help in the profound delivery of the bio-fertilizer inside the plant [36–40]. CNTs are used for delivering biofertilizer. CNTs facilitate nutrients delivery, which is helpful in humanizing their growth and yields [41–43].

Carbon nanotubes (CNTs), because of their distinctive 1D tubular structure and other inimitable properties such as electrical, thermal conductivity, surface area, etc. have been successfully used for improving the electrochemical characteristics of both the anode and cathode of Li-ion batteries. Various reviews are available in the literature which explains CNT-based composites are widely used for enrich energy conversion and storage capacities, and power density, capability, life cyclic, and protection of LIBs [44].

2. Carbon Nanotubes for Water Purification

Water is an important natural resource required for the existence of human beings. Human life cannot flourish without the availability of clean water. Water is a nonrenewable resource. Water contains different pollutants which makes it unfit for drinking water. Water is used in all scats of life. Different recent technologies are available for cleaning water is not as much effective. Carbon nanotubes are able to remove numerous types of pollutants in organic, inorganic, or biological materials. Lots of waste water is generated by the industry, which creates an ample range of landfills. Carbon nanotubes are able to clean industrial wastewater which in turn reduces landfills and protects our environment [45]. The carbon nanotubes based on specific nanotechnologies discussed in the paper solely focus on site remediation and some waste water treatment, for example: they have been recognized for their ability to absorb dioxins much more strongly than traditional activated carbon. They act as excellent adsorbents for the removal of numerous pollutants, including microorganisms from the drinking and water bodies present on earth. They have various magical and unique properties, which make it versatile such as excellent thermal stability, higher adsorption ability, broad range of pH, etc [45].

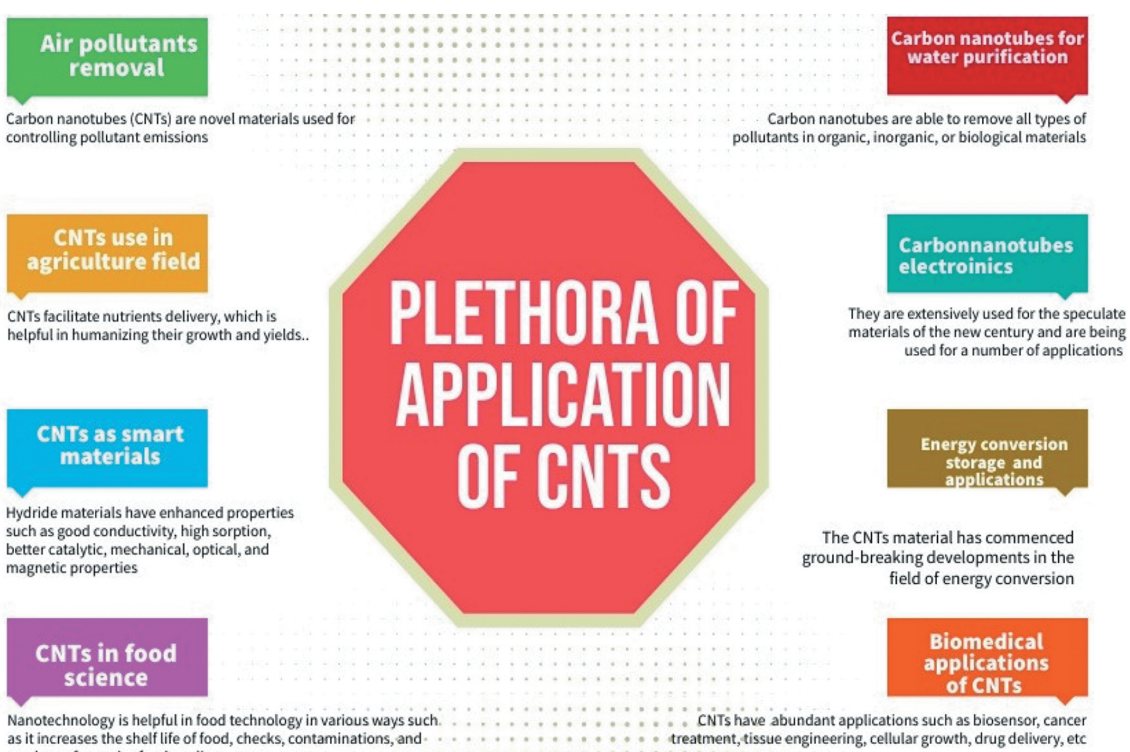

Figure 2. Plethora of applications of carbon nanotubes in various fields such as textile, biomedical, health care, food industry, electronics, environment, renewable energy, etc.

Carbon nanotubes have been utilized for the elimination of various harmful toxic substances from drinking and other wastewater, because of their well-defined structure and adjustable characteristics. Numerous researchers studied the strong exchanges between CNTs and heavy metal ions and with organic compounds. This is because of the surface functional groups and hydrophobic surface of CNT [46]. Modifications can be made in carbon nanotubes to improve its selectivity and removal of metals or ions from aqueous stream [47].

Functionalization of carbon nanotubes may also act as a promising option for the removal of water pollutants. Carbon nanotubes are used to remove several pollutants from water bodies which are harmful for the environment. Such compounds are organic in nature, pesticides, pharmaceuticals, antibiotics, and inorganic ions such as heavy metal ions [48].

Due to the structure and morphology of the carbon nanotube, it is useful for the adsorption of various organic and inorganic pollutants. They are far better than activated carbons in terms of their lesser equilibrium time, higher adsorption capacity and selectivity, and easier regeneration.

Nanocomposites such as magnetic polymer multiwall carbon nanotubes (MPMWCNTs) are formed by the combination of poly (1-glycidyl-3-methylimidazolium chloride) (an ionic liquid-based polyether) and ferroferric oxide. Such nanocomposite is very effective to remove various dyes such as orange II dye, sunset yellow dye, FCF, and amaranth dye from aqueous phase [49].

MPMWCNT is also effective in removing dye-containing wastewater because of high adsorption capacity, fast adsorption rate, and it separates from water if an external magnetic field is applied on it. The adsorption of ofloxacin (OFL) on carbon nanotubes (CNTs) has been reported by researchers and scientists by means of solubility, pH, and cosolvent effects [50] carbon nanotubes (CNTs) are cost-effective, simplistic, green, and extremely effective real/fruitful in water and wastewater treatment. Figure 2 illustrates the applications of carbon nanotubes in water purification.

Nano-structured TiO_2 are present in size, shape, and structure (particles, fibers, wires, rods, porous materials, and tubes all in nano dimensions). These morphologies are prepared and used in several applications [51]. Tubes of titanium dioxide have varied applications in the environmental field. After carbon nanotubes, TDN is the second most abundant. TDNs synthesized from the anodization of metal plates have substantial applications in water photo-electrolysis, hetero-junction, solar cells, gas sensing, and environmental purification [52,53], as they provide large surface area. Metal impurities in carbon nanotubes are accountable for the 'electrocatalysis' found at nanotube-modified electrodes. Metal inclusions may have a role in the reported 'electrocatalysis' linked to carbon nanotube-modified electrodes in some cases. Edge-plane pyrolytic graphite (EPPG) electrodes should be compared to carbon nanotube-modified electrodes in all electrochemical investigations. It will either reveal that edge-plane sites/defects are to blame, or it will imply that metal impurities may facilitate an electrochemical reaction.

The TDN provided is their inner diameter of approximately *101 nm* and an average length of around *3.5 μm*. Order structure of the TiO_2 film gives a unidirectional electrical channel and large surface area, which enhances photo-catalysis and photo-electrochemical applications [54,55].

Electrochemical oxidation of methyl orange degradation reported by Zhang can be possible effectively when combined with UV irradiation. Percentage will be *56.3%* in *90 minutes* where an initial dye concentration of 2×10^{-4} *M* [11]. TiO_2 nanotubes are used in electrocatalysis and act by electro-activating materials like noble metals (Ag, Au, Pd, Pt, and Ru) or metal oxides. TiO_2 is also used in electrochemistry because of semiconductive properties, longer stability [56]. Gold, silver, palladium, platinum, and Ruthenium nanoparticles (*3 nm*) loaded on TDN layers enhance TiO_2 PC properties [57].

3. Carbon Nanotubes in Electrical/Electronic Applications

Carbon nanotubes have the symmetry and unique electronic structure of graphene; nanotubes have a very high current carrying capacity. They are extensively used for speculate materials of the new century and are being used for a number of applications [58] ranging from automobiles to nanometer scale electronics. The electrical properties are useful for saving energy and in turn decrease the use of wood cutting and the use of traditional batteries by providing better electrical materials for future generations [59,60]. Carbon nanotubes provide small size, carbon-based nanoelectronics having high flexibility compared to conservative silicon electronics. Carbon nanotubes have unusual properties because of the low dimensionality [61]. Carbon nanotubes can be used to harness electronics devices, which is helpful in power saving, radiation, hardness, and reduced heat dissipation. Carbon nanotubes can be effectively used to relate the phenomena of photoconductivity, thermoelectricity, and superconductivity [61]. The basic conducting features of a graphene tubule are also known to be dependent on the type of wrapping (chirality) and the diameter (usually, SWNTs possess diameters ranging from 0.4 nm to 2 nm). The CNTs replace traditional batteries such as electrochemical capacitors or super capacitors, which are not only given their miniature size, elevated power density, prolonged for a longer period of time, and high energy density, which decrease the waste disposal to the environment.

3.1. CNTs as Super Capacitors

Super capacitors consist of large surface area activated capacitors, which are having molecule-thin layer of electrolyte as dielectric [62]. The CNT is also used as an electrode material for the capacitor to augment the ability of the capacitor to store higher energy density because of the large surface area [63]. The CNTs possess high surface area, stability, and strong mechanical properties. The CNTs are not acting stronger candidates with low capacitance [64]. Therefore, CNTs, but also used actively in good specific capacitance transition metal oxides, for example, manganese oxide (MnO_2) and ruthenium oxide (RuO_2) [65,66]. The CNTs provide capacitances of approximately *5000 F* and energy densities up to *5 Wh/kg*, which is higher up to 10-fold higher than conventional capacitors, with only *0.5 Wh/kg* [67].

3.2 CNTs material has commenced revolutionary developments in the field of energy conversion, storage, and applications

Humanity is facing many issues out of top problems. Energy and environment both are in a prioritized list. Renewable sources of energy like oil, gas, and coal are fulfilling the need of energy until now, but they are depleted and exhausted because of the overuse. Pollution produced by these fossil fuels are numerous, because they increase greenhouse gases, global warming, and in turn is a major cause of natural disasters and environmental degradation. Industrial revolution, agricultural activities, metallurgical activities, and application of pesticides and herbicides require energy for sustainability.

Growth in any sector requires energy. Energy generated by conventional sources leads to pollution. Contamination of air, water, and soil leads to the depletion of our environmental quality. Quenching the requirement of energy for the protection of the environment is utmost important. Nanotechnology is now addressing the problem of energy by giving substituted products which are more efficient and have smaller dimensions showing new properties and behavior than bulk materials.

The CNTs have very distinctive structural properties and show unique electronic, chemical, and mechanical properties. The main culprit of global warming effects is fossil soil, which has been exploited for years. The CNTs based material has commenced ground-breaking developments in the field of energy conversion, which includes the development of solar cells, fuel cells. It also brings new developments in the field of energy storage such as hydrogen storage, lithium ion batteries, and electrochemical super capacitors, the hybrid nanocomposites of CNTs have also developed to save energy [58].

4. Carbon Nanotubes as Electrochemical And Electronic Biosensors

The carbon nanotubes used in biosensors are extremely successful and outstanding due to its exceptional properties such as fast electron transfer rate [68] and high electron transfer helping in good conductivity, better aspect ratio, elevated conductivity, good chemical stability and better sensitivity [69]. The basic requirement of biosensors is the control of biomolecules with its surface, which helps in the detection and signal transduction process. The CNTs improve electron transfer which is very useful in combining electrochemical and electronic biosensors [70,71]. Electrochemical biosensors developed on CNTs are used for distinguish nitric oxide and sensing epinephrine [72,73]. Several Scientists such as Bisker et al. 2016 [74] used the CNTs for detecting human blood proteins using 20 distinct SWCNT corona phases. The MWCNTs based device has been developed by Scientists Baldo et al. 2016 for detecting arginase-1. The SWCNTs are severed as operative optical probes in biomedicine due to their unique photophysical properties, good fluorescence emission in the NIR region, and photo stability [75].

5. Biomedical applications of CNTs in various fields

Medical field is mainly governed by good drug administration. Due to less selectivity and low half-life time, the drug cannot fulfill the real purpose of the treatment. To address the problem of multiplying the administration of drugs to fulfill the needs of the body, this leads to fatal issues. The CNTs are extensively used as nano carriers for drugs, gene delivery and cancer treatment and other biomedical applications. CNTs are extensively used and exploited in various biomedical applications such as biosensing, drug delivery, and cancer therapy. Presently, in this review we carefully reiterate the development of CNTs for biomedical applications [76].

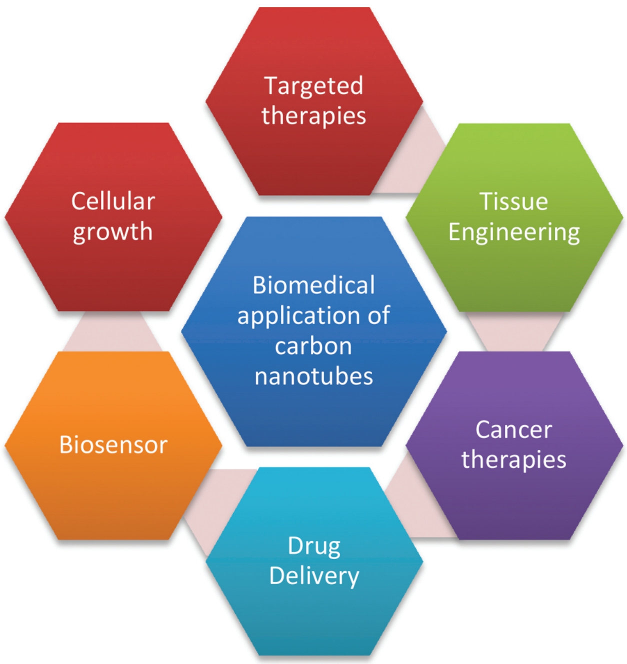

Figure 3. The figure represents the biomedical applications of CNTs such as targeted therapy, cellular growth, tissue engineering, cancer therapy, drug delivery, biosensors, etc.

CNT is used for targeted therapies such as drugs, gene delivery and cancer treatment and other biomedical applications. It has been seen that carbon nanotubes have numerous applications in various fields such as removal of water and air pollution, energy storage, semiconductors, food science, etc. However, the most important application is biomedical applications which safe the life of the human. Out of abundant applications such as biosensor, cancer treatment, tissue engineering, cellular growth, drug delivery, etc. Most significant is cardiovascular disease. India is considered as the hub for cardiovascular diseases. In 2016, the estimated dominance for cardiovascular diseases in India was approximate to be *54.5 million*. Every fourth death is caused by heart disease, out of which *>80%* heart disease is due to stroke [71].

5.1. Carbon Nanotubes for Drug Delivery and Cancer Therapy

CNT has gained attention in recent years because of its ability to attach to drug molecules via various types of bonding or functionalized CNTs and deliver drugs into living cells. Considerable attention was gained by CNTs because of noninvasive penetration across biological membranes [77,78]. CNTs act as a nano carrier in the last recent years, which shows exceptional cell transfection capabilities. Several researchers such as Liu et al. loaded the anticancer drug doxorubicin (DOX) with poly (ethylene glycol) (PEG) conjugated SWCNT which reaches to timorous tissue [79,80]. Some more examples are given which is covalently linked to MWCNT anticancer drug, 10-hydroxycamptothecin (HCPT) [81]. The CNTs are good in intracellular drug delivery and gene and protein delivery. Xu et al. also wrought magnetically directed nano carriers consuming a paramagnetic surfactant coating [82]. By seeing these examples, it can be concluded that CNTs are effectively used in drug delivery systems and act as nano carriers. Versatile properties of carbon nanotubes make it suitable for biomedical applications. They are used in delivering drugs for anticancer drugs, genes, and proteins [76]. SWCNTs are a kind of SP^2 hybridized carbon that looks a lot like fullerenes. The hexagonal composition of the structure is similar to that of graphite, forming six-atom carbon rings. MCNTs, on the other side, are made up of multiple tubes arranged in concentric rings in a cylindrical structure.

5.2. Potential Applications of CNTs in Tissue Engineering and Cellular Growth

Brain is a part of the body which to be delicately handled. Growth of the brain cell cannot be possible on any of the media. CNTs have been used as a promising material for neural signal transmission and it provides a good surface for cellular growth.

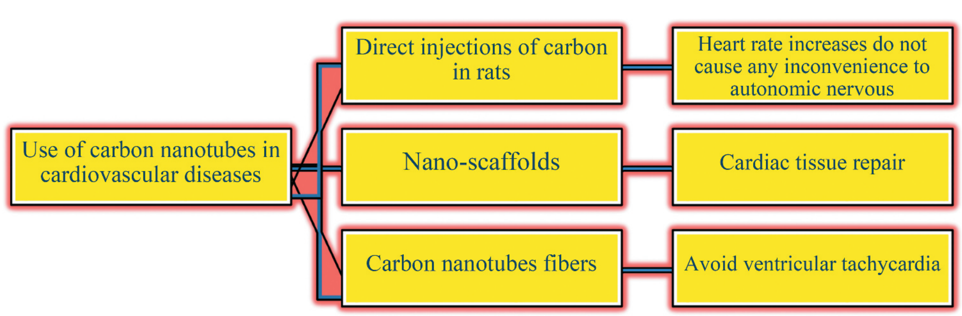

Figure 4. The figure represents the use of cardiovascular disease.

Several researches and scientists such as Mazzatenta et al. worked and concluded that SWCNTs stimulates brain circuit activity when SWCNTs and hippocampal cell combines [83]. Neurons would better grow on double walled carbon nanotubes (DWCNTs) because of the surface texture and good adsorption of culture medium than on SiO_2 surface, which helps in protein tolerating the growth of neuron networks [84]. Figure 3 represents the biomedical applications of CNTs.

5.3. Biomedical Importance of CNTs in Cardiovascular Diseases

Carbon nanotubes (CNTs) are novel materials which could save the world from cardiovascular diseases. It is widely used in various cardiovascular diseases. The nano-compatibility of the carbon nanotube proofs it as a novel candidate for future advance research in the field of medicine and biology.

Indeed, it is a fact that carbon nanotubes (CNTs) have a brilliant prospect in the field of revolutionary product research [85]. There are certain ways in which carbon nanotubes are used as they are represented schematically in Figure 4.

5.4. Cardiac Effects of MWCNTs by Direct Injection in the Rat's Body

The investigation of the cardiac effect of MWCNTs by direct injection has been seen in rats. When the CNTs are directly injected in rats, it was found through electrocardiogram (ECG) signals that the heart rate increases significantly. The CNTs do not cause any inconvenience to the autonomic nervous system (ANS), However scientists have to work with biocompatible nanotubes in the heart [85–89]. The CNTs have established that they have a strong ability to act as scaffolds to endorse cardiomyocyte propagation, maturation, and survival [41].

The CNTs fibers have the ability to transfer electrical signals to the myocardium. The CNTs provide electrical conductivity, strength, and flexibility to facilitate signals from blood vessels. They improve the conduction of ventricular and atrioventricular blood vessels. They provide uplifting solution to avoid ventricular tachycardia [42,43,90,91].

Furthermore, there is a constant need to check the biocompatibility of the CNTs inside the body. Many researchers are working hard in the field of materials-based improvements. They are facing many challenges to improve atrioventricular conduction (a pacemaker suffices), which is the main reason for the intramyocardial delay. Integration of the neuroscience surface modification is the need of the hour. In vivo reconstitution should be studied further for the biocompatibility of CNTs. However, the positive results of the CNTs based materials send new ray of light which is helpful in avoiding sudden death. After the survey of the literature, it is impossible to overlook CNTs based materials are having vivid applications in the biomedical field. CNTs can be used for practical applications on sensitive parts of the body [91]. Since, the development of biocompatibility of CNTs and surface improvement are required. Therefore, to preserve the balance in the structure of CNTs and modifying the CNTs to contemporary needs is still a challenging task. Moreover, it is likewise essential to improve the biocompatibility of CNTs while retaining natural properties. The review can also motivate researchers to explore improvements in various arenas of biomedical, environment, energy storage, semiconductors, food science, etc.

6. Air Pollutants Removal by Using Carbon Nanotube Filters

The impact of air pollution on human and other animals, plants living on earth is large. Greenhouse gases present in the environment make the mother earth hot. Greenhouse gases are the major reason for global warming. Greenhouse gases such as water vapor, carbon dioxide, methane, ozone, nitrous oxide, chlorofluorocarbons, etc. are the major contributors. In urban areas, diesel and gasoline fuels are mostly used as a nonrenewable source of energy, which is the major contributor of air pollution. Incomplete combustion of fuel emitted from automobiles and vehicles are the major contributor. Nitrogen oxides as NO combine with the ozone layer and deplete the ozone layer.

A volatile organic compound (VOCs) contains the organic compound benzene, which is very toxic for living organisms. Volatile organic compounds (VOCs) are generated by fuel combustion and incomplete combustion. Epidemiologic studies have reported that cardiopulmonary diseases and lung cancer are caused by air pollution. Daily increases in air pollution in cities, which are the major contributor of deaths [92]. Air purification is a subject of interest for environmentalists. They are looking for substitute technologies to overcome the conditions that happen seriously in human life and for the entire biosphere. Sorption process for storage and purification are used for decades. Carbon nanotubes (CNTs) are novel materials used for controlling pollutant emissions because of the unique chemical and physical properties. Carbon nanotubes (CNTs) are used as sorbents for controlling environment pollutions [93]. Carbon nanotubes (CNTs) are developed on membrane support of diverse materials with pore sizes of around *10.5 mm*. CNTs are used to remove gases harmful and competence depends on diverse variants. Varied range of new filters has been developed using different materials such as metal or ceramic trusses. The CNTs based filters developed on Si/SiO_2 stainless steel or glass fiber show good filter efficiency for various particle sizes [94–96].

Various materials originated from carbon, for example, active carbon and carbon fiber, have extensively used in the gas sorption processes. In the current years, CNTs are extensively used for controlling pollutant emissions and they are acting as very good sorbent for environmental applications. Nanotube filter was efficaciously produced and exploited to remove bioaerosols and acted well with biological agents from the water for filtration [93,97–106].

7. Cases and Studies on the Toxicity Through CNT

Nano sized particles as being naturally present in the environment for millennia in the form of viruses, volcanic ash, and forest fire ash, etc. Human exposure to nano-sized particles leads to health-related problem. Industrial Revolution has increased the count of nano-sized particles noticeably. Polymer fumes, fumes produced by burning of metals, incinerators, or combustion engines. Nanoparticles are produced naturally as byproducts from these reactions. Growth of nanotechnology applications in various fields and exposure of nanoparticles to human and living beings is enviable. These exposures create several health-related issues in the living being. Health related should be taken care when there is use of nanomaterials such as CNTs.

7.1. Water soluble Carbon Nanotubes Increase Agricultural Yields

Agriculture sector is facing difficulties because of the lack of advanced technologies. Boost in agricultural production could be possible using nanotechnology-based applications. Nanotechnology is useful in controlling the delivery of nutrients which in turn increases the crop production. High-tech farming is supported by their direct and indirect applications of nanomaterials. When the plants are able to absorb a good amount of nutrients from the soil, the agricultural production is increased [107,108]. Studies had proved that CNTs play a supporting role in both in vitro and in vivo. Both types of CNTs such as MWCNTs and SWCNTs are useful in delivering nutrients to plants as they are able to cross the plant cell walls and reach cellular organelles. MWCNTs are shown to be much stronger than SWCNTs in experiments, Water solubility of CNTs is increased by chemical modification [109]. Chemical modification can be possible by introducing a variety of functional groups (covalently and non-covalently) on their outer surfaces. Some of the methods for the modifications in the gas phase are ozone-induced oxidation of CNTs; the second known methods are UV-ozone oxidative dielectric barrier discharge.

Internal translocation means a chromosome is transferred to another chromosome using CNTs. CNTs are used for delivering biofertilizer. When biofertilizer is delivered through CNTs, the unique properties of CNTs will provide large surface area, superior reactivity, and smaller particle size, which support the profound delivery of biofertilizer inside the plant [110]. The CNTs facilitate nutrients delivery, which is helpful in humanizing their growth and yields. When CNTs are administered at low concentrations, the positive response is shown by the plants. However, if the concentration of CNTs is more, then sometimes it causes damaging effects on plant growth. The CNTs effects on the surroundings should be investigated before extensive use in agriculture. Protection of plants should be on a priority base before large-scale use of CNTs is there [111].

7.2. Role of Carbon Nanotubes in Reducing Environmental Pollution

Carbon nanotubes (CNTs) play an imperative role in the removal of environmental pollutants. Protection of the environment and health is the major concern for scientists and environmentalists in the route of protecting the environment [1]. Nanotechnology has a broader perspective for protecting environmental health and sustainable development. The scientific data has been collected from various sources such as Google Scholar, Scopus, Elsevier, and PubMed and analyzed to evaluate the results. Out of the reviewed data, it was concluded that general methods of sewage treatment are not adequate to remove the environmental pollutants totally. The CNTs used adsorption methods are simple and require less energy consumption in comparison with conventional methods. It is extensively used in various applications such as drinking and sewage water treatment, and in removing pharmaceutical pollutants, in the treatment of leachate, in the removal of petroleum compounds, heavy metals, in dye removal, and in removing leachate contaminants [2,3].

With the increase in urbanization and population exploitation, there is a constant need for the protection of the environment for long-term survival of human beings. Controlling pollution generated by human

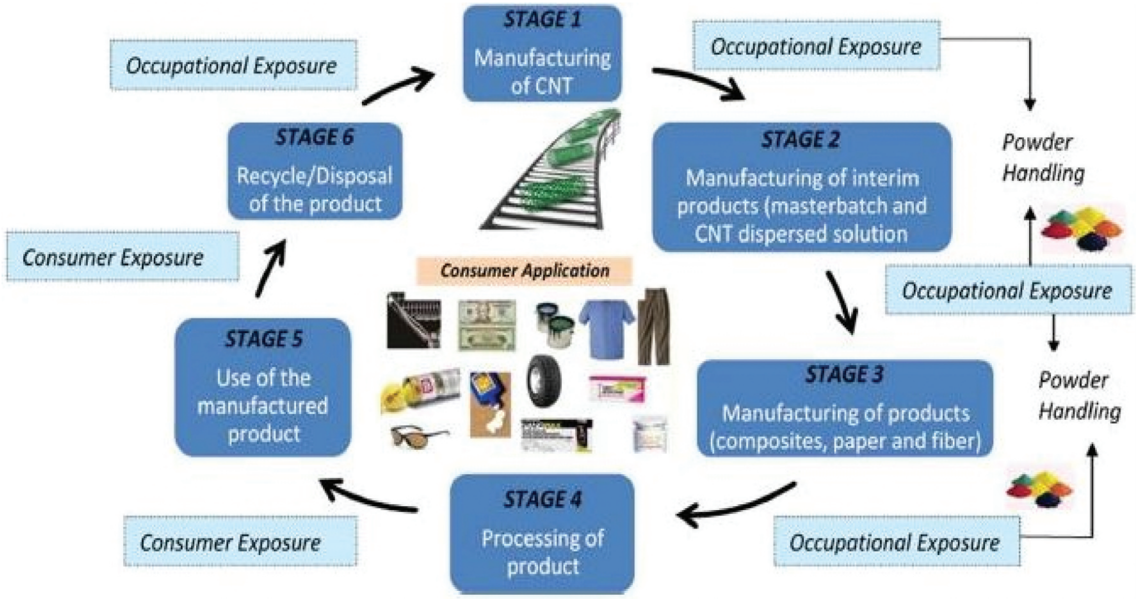

Figure 5. The life cycle of CNTs is related to risk assessment studies in which the first, second, third, and sixth are the stages of occupational exposure. The fourth and fifth stages come under consumer exposure (Reprinted with permission from Ref [11]. Copyright 2018 Springer Nature).

intervention is needed on the priority base. Easy, fast, high problem solving, cheap technologies are in demand. The research can be concluded as effective in establishing an impression that CNTs as effective adsorbents, have high prospective in removing pollutants which helps in sustainable development. It can be reused, recycled, and reduced consumption of fossil fuel. CNTs are painstaking eco-friendly materials which reduce carbon footprint in the environment [4,5].

7.3. The Toxic Effects of Carbon Nanotubes in Water Purification

The elongated-term use of carbon nanotubes (CNTs) for water purification requires nanosafety guidelines. Risk measurements decrease the toxic effects of CNTs. Different purification methods of waste water treatment by CNTs involve certain types of risk, which need to be studied through risk assessment of CNTs [11]. Scarcity of knowledge in terms of health risks associated with CNTs made scientists to design risk estimates. There should be universal CNT safety guidelines for analyzing the CNTs risk [12]. To narrow down the gaps, various new risk analysis routes and CNT-based water purification technologies have been developed to optimize the results. The CNTs safety clock has been designed to assess risk evaluation and organization. The CNTs safety guidelines have been designed by taking concerns about the diameter of CNTs, its length, aspect ratio, charge, hydrophobicity, functionality, etc. It helps in establishing CNTs actions in waste water treatment plants and consequent discharge into the environment [13].

In Figure 5, the life cycles of the CNTs, in brief has been discussed in six stages. The first, second, third, and sixth are the stages of occupational exposure. The fourth and fifth stages come under consumer exposure. The first stage in which the CNTs are manufactured occurs in a closed furnace without oxygen intrusion to avoid exposure to CNTs. The master batches and CNT-dispersed solutions are the intervening products that have been synthesized in the second stage. In the third is the product formation stage in which the handling of the CNTs will reduce. The fourth and fifth stage is the processing of products and final use of the product. And the last stage is disposal or recycling [14].

Many safety issues, risk assessment, and control measurements for E-CNT (Environmental CNTs) have been discussed in the research paper. With diverse applications, different types of risk are also a concern for the scientist. Application-specific risk assessment is utmost important in the case of nanomaterials such as CNTs. Risk associated with the use of nanomaterials creates a type of nano phobia among people. Scientists and researchers are working hard to remove such types of nano phobia by providing safe water facilities, conserving our environment.

The CNTs in water purification require proper handling, utilization, and discarding of nanomaterials in a more precise way. To address the safety challenges

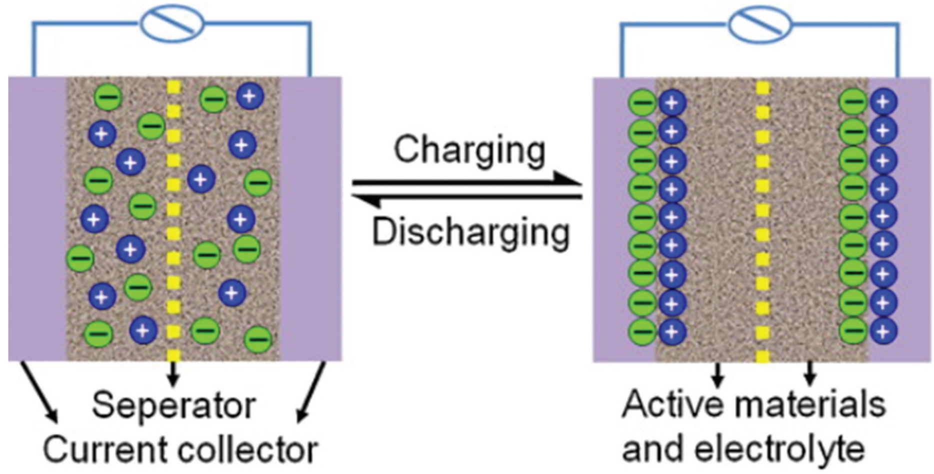

Figure 6. Schematic diagram of a super capacitor cell shown here is composed of high surface area activated capacitors with a very emaciated layer of electrolyte as dielectric. The CNTs act as superior electrode materials for capacitors (Reprinted with permission from Ref [115]. Copyright 2013 Elsevier).

associated with nanomaterials, there is a constant need of safety guidelines and regulatory frameworks. These safety guidelines and regulatory frameworks not only categorize and differentiate nanomaterials but also assess the impact on human health and the environment. Case-to-case study measures the profound risk assessment of nanomaterials' structure–property relationships. The assessment helps in fixing the universal safety guideline which accurately keeps an eye on the prospects of the CNTs [112].

7.4. An Insight into Sustainable Applications of Carbon Nanotubes in Protecting the Environment

In line with the sustainable environment and green technology perspectives, carbon nanotubes (CNTs) act as a promising material for recent developments. The CNTs, due to its distinctive properties, are set to be a benchmark in the field of pollution control and monitoring, waste management, composites, water treatment, biotechnology, super capacitors, renewable energy etc. and act as popularly known materials for decades. The CNTs have astonishing physical, chemical, and electronic properties which act as cost-effective in various applications. The potential hazards of these nano materials have been discussed in the review. CNTs are wonderful materials used in various environmental fields which come to know from the literature [113,114].

Reduction of waste disposal in the environment can be achieved by using the CNTs as super capacitors which provide small size, high portability, utmost power density, extended suitability, and elevated energy. Figure 6 shown here is composed of maximum surface area activated capacitors with a very emaciated layer of electrolyte as dielectric. The CNTs act as superior electrode materials for capacitors. Generally, the CNTs are coupled to a thinner layer of the electrode, increasing the capacity of the capacitor to accumulate higher energy density. The CNTs are placed in between the electrode and electrolyte. Vertical alignment of the CNTs increases the capacity of the super capacitor by increasing the surface area of the electrode. The CNTs alone are not suitable to use the electrode due to their little capacitance. However, CNTs combines with transition metal oxides to produce sustainable energy. The metal oxides are such as manganese oxide (MnO_2) and ruthenium oxide (RuO_2), etc. The energy density *(5 Wh/kg)* up to 10 folds increased using capacitors, with only around *1.0 Wh/kg* [115].

The green energy produced by clean combustion is the need of the hour. Biofuel cells created by biomass catalytic activity, photovoltaic devices formed from renewable sources such as solar, hydrogen fuel cells using hydrogen as feed produces energy without producing toxic gases to the atmosphere. These alternative energy producing devices prohibit the elevated demand for fossil fuel. On the other hand, these technologies are still required to cross many roads to fulfill growing energy demands. These technologies are on the emerging stage of development, practically needing more commercialization. The CNTs-based super capacitors produced superior power density bestowed as an

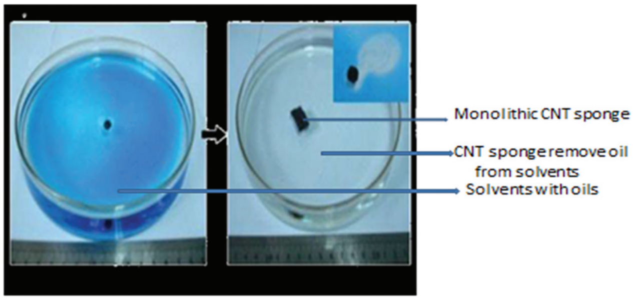

Figure 7. Application of CNT sponge in the cleanup of oil from water which shows a monolithic CNT sponge prepared by chemical vapor deposition via ferrocene as a precursor (Reprinted with permission from Ref [118]. Copyright 2015 Premier Publisher).

alternative fueling replacing traditional batteries. Super capacitors have high capacity and extended usage time that reduces the generation of waste in the environment. The involvement of CNTs as a nanocomposite design which is green in nature can fulfill the 3 R (reduction, reuse, and recycle) capability of raw materials. Low cost of CNTs also favors the usage of CNTs in various fields. Source reduction of CNTs should be avoided for safe human health hazards of CNTs [116,117]. It also demonstrates that MWCNTs perform even easier in corrosive environments. When used as a catalyst for fuel cells, MWCNTs exhibit lower loss or oxygen depletion than SWCNTs.

7.5. Environmental Applications of Carbon Nanotubes

Carbon nanotubes are the most studied materials in modern times. The CNTs have astonishing physical, electronic, and chemical properties, which creates an extensive diversity of applications. Here, the contribution of CNTs in various fields has been discussed, such as air pollution, water pollution, and safe renewable energy, developed energy, super capacitors, and green nanocomposites, etc [118]. It also works extensively in the field of biotechnology, providing sustainable environment and green technologies [119]. For deliberation of CNTs at a large scale, we need to discuss various aspects of charge and impending hazards. Based on the studies, it could be easily given the impression that CNTs have immense potential and are capable materials for large-scale application in various environmental fields [120].

Figure 7 shows a monolithic CNT sponge prepared by chemical vapor deposition via ferrocene as precursor. The CNTs sponges have elevated porosity and very little density and have haphazardly entangled three-dimensionally.

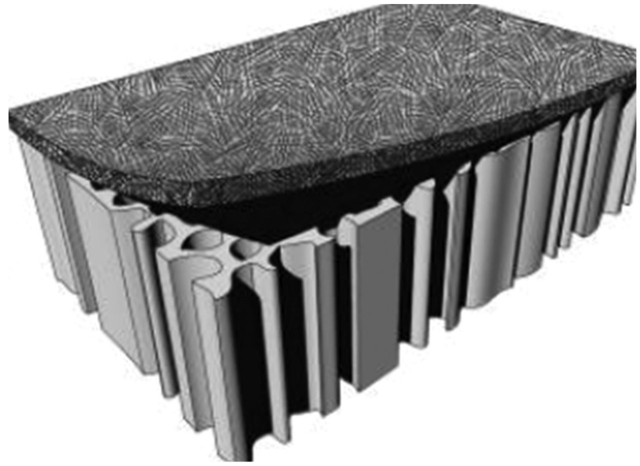

Figure 8. Hybrid filter for effective water purification in which a porous ceramic supports will provide the CNTs cross-linked filtration layer (Reprinted with permission from Ref [123]. Copyright 2008 American Chemical Society).

The CNTs sponge is useful in removing oil in large quantities, i.e., *80 to 180 times* their own weight, and can effectively be used in an extensive range of solvents and oils. The CNTs sponges have high floating-and-cleaning capability because of high hydrophobicity. Regeneration of CNTs sponge is easily possible by mechanical compression. Waste products could be easily burned in air without destroying the sponge structure [121].

The CNTs have been established as a large amount of practical and effectual material in diverse applications. It helps to create a simple process out of the complicated one which reduces the usage of material and resources which in turn decrease the exploitation of resources. Apart from this, the potential application of materials, health hazards, and production is also an area of concern. It covers all areas right from pollution control, environmental engineering, and biotechnology. It is very well said that nonmaterial would rule the world in technological development [122].

7.6. Future Applications of Carbon-Based Nanomaterials in Protecting the Natural Environment

Environmental challenges make human beings in search of new materials to solve their problems. Carbon-based nanomaterials are promising materials that address all challenges related to the environment [123]. The review touched out the arenas of environmental applications, such as water purification, membrane sorbents, environment sensors, and different modified filters for water and air purification, electronic materials which in turn helpful to save the renewable energy and prevent pollution [15]. In view of the unique physical, chemical, and electronic properties of the CNTs, the present article fully explains the jest of future opportunities of nanomaterials' in protecting the environment.

Hybrid SWNTs ceramic filters have been developed and open new avenues in this field. They are having tapered diameter, large surface area, and good thermal resistance which combines the porous aggregate structure of SWNTs. In Figure 8, a porous ceramic support can provide the CNTs cross-linked filtration layer. These layers are highly permeable SWNTs filters demonstrated good efficiency in removing biological disease causing organisms from water. They are demonstrated elevated resistant to high temperatures and adequately durable for repeated reuse in water treatment applications [16].

Incorporation of green technology in product design and manufacturing is utmost important for saving our environment. Various opportunities for product design; regeneration for secondary applications has been explored. Nanomaterials which are used once could find it potential application in the form of 'Best out of waste' in various other fields like structural reinforcement or road construction. To minimize the disposal of nanomaterial's in the environment, certain types of strategies should be fixed. The multidisciplinary collaboration may act as a gift for future generations, which is possible by the joint effort of the product development engineer, environmental scientists, and toxicologists [17]. Several challenges act as barriers in this road for scaling up the technology from lab to industry. High price availability of nanomaterials is always a challenge in coming times. Waste management in the existing infrastructure system according to the present need is not sufficient to provide lodging for new nanomaterial's products. The risk of these new technologies is always a matter of concern for the scientist. Public awareness and concern about the risk of nanomaterials is also screwing up the minds of stake holders. Efforts are made to earn people approval and enhance the arena for nanomaterials, which is helpful in creating a conducive environment [18].

7.7. Life Cycle Assessment of Semiconductor Devices with Carbon Nanotubes

The use of carbon nanotubes (CNTs) in electronic applications is very much rampant due to its inimitable mechanical, thermal, and electrical properties. The environmental impact and health hazards of carbon nanotubes (CNTs) have been a concern with the increase in the commercialization of nanomaterials. There is should be short of the rate of information concerning the potential health hazards and environmental impact caused by the CNTs. The study of life sequence environmental impacts of CNT helps to evaluate the possible adverse effects caused by CNTs [19]. It enhances and motivates scientist researchers to develop

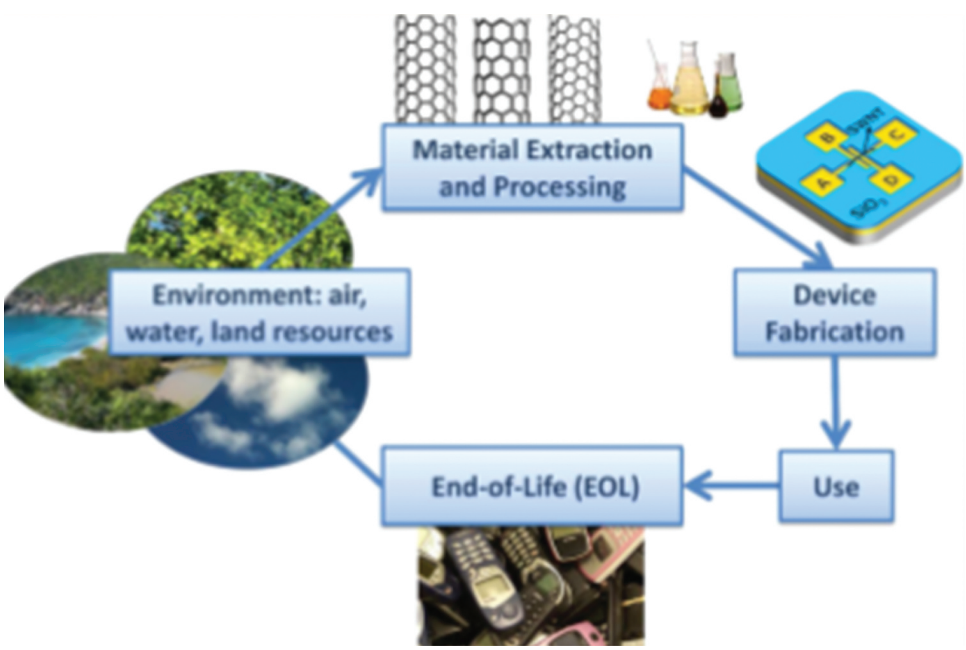

Figure 9. The figure shows life cycle assessment of semiconductor devices with carbon nanotubes. The present figure estimates the impending application of CNT switches to contemporary cellular phone flash memory (Reprinted with permission from Ref [10]. Copyright 2013 American Chemical Society).

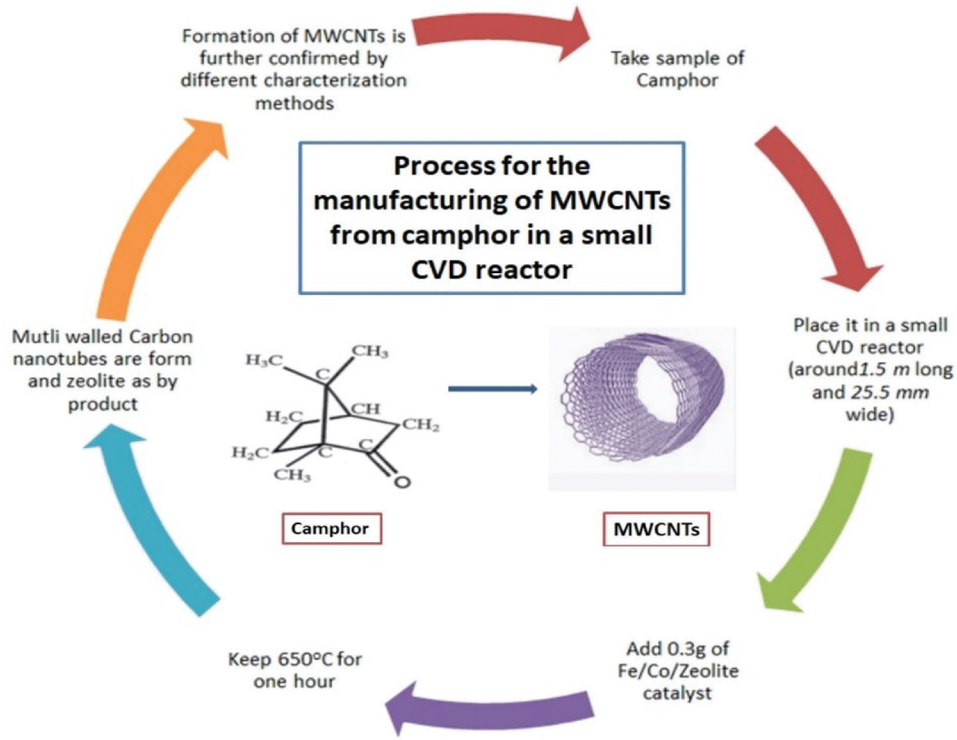

Figure 10. Step wise process for the manufacturing of MWCNTs from camphor in a small CVD reactor.

new products which minimize the adverse effects of CNTs.

The present Figure 9 estimates the impending purpose of CNT switches on contemporary cellular phone flash memory. Life cycle assessment (LCA) is an important tool to assess the environmental flow of nonvolatile bistable electromechanical CNT switches by studying the stages right from the beginning of fabrication application and end-of-life [20]. The study of the life cycle from first-order anticipated use of the CNTs product and end-of-life (EOL) stages indicates that there is a constant need of reducing the environmental burden at every stage.

The CNTs switch technology could be replaced by using NAND flash FGMOSFETs, which provides better improved results and in turn decreases the environment landfill and circumvents the release of CNTs. EOL (end-of-life) management of the CNTs switch designated in the separation and metallurgical processes used in the present day needs to be modified for proper release of the CNTs in the environment [21]. Lots of efforts need to be done at the modeling level to measure, control, and optimize the release of emissions from nanomaterials starting from the fabrication to the end of life level. There is always a need of qualifying each stage of the life cycle for better environment sustainability [22]. The fact is that CNTs technology is useful in reducing the energy expenditure and increasing usage in lots of applications. Knowledge of the life cycle will help scientists to start from the fabrication level through the development

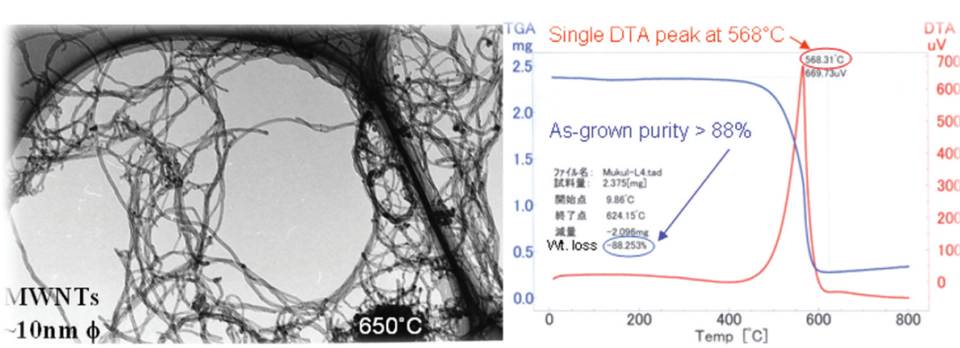

Figure 11.

stage until the final stage. The negative environmental impact is reduced by EOL management and brightens the future of CNTs-enabled technologies [23,24].

7.8. Potential Applications of Promising Carbon Nanotubes Prepared Using Organic Camphor

Environment friendly process is the need of the hour. Saving the environment is the paramount task for the human race. The synthesis technique depicted here fully addresses the '12-principle' practices of green chemistry. Camphor is known as an environment-friendly hydrocarbon. Elevated-purity carbon nanotubes (CNTs) can be produced from camphor hydrocarbons using a chemical vapor deposition method. The process can be accomplished in a small chemical vapor deposition (CVD) reactor (around *1.5 m* long and around *25.5 mm* wide). Approximately yield of *1.62 g* multiwalled carbon nanotubes (MWCNTs) of diameter *~10 nm* can be obtained by *3 g* camphor by optimizing the temperature of *650°C* at *1 h*. Generally, the camphor-to-CNTs manufacture efficiency is *50%*. However, optimizing the process efficiency would be increased up to *88%* [124].

Figure 10 clearly depicts the manufacturing of camphor in a small CVD reactor (around*1.5 m* long and *25.5 mm* wide). The process requires around *3.5 g* camphor which is vaporized at around 201°C under a tolerant argon flow of around *51 cm*. It is further pyrolyzed at around *0.35 g* Iron cobalt saturated zeolite support at around 655°C. The process can blow up the zeolite bed more than fifty times by volume, this leads to help in obtaining the yields of around *1.63 g* MWCNTs of diameter ~10 nm. The purity is further increased by *88%*, which is clearly demonstrated in the characterization process shown by transmission electron microscopy (TEM) images, and thermogravimetric analysis (TGA).

Figure 11 **clearly demonstrates the characterization process with TEM (transmission electron microscopy) image and TGA (thermogravimetric analysis) curves of camphor and zeolite beads (Reprinted with permission from Ref** [124]. **Copyright 2007 IOP Publishing)**

Nanotechnology opens all avenues and is widely used in all applications. As we all know that the carbon nanotubes (CNTs) are single-layer carbon atoms (graphene) and are rolled-up sheets in a cylindrical manner. Carbon nanotubes (CNTs) have incredible potential in nanotechnology and a gigantic amount of effort has been invested in producing CNTs worldwide [125–127]. The above technique fully is in line with the 12-principles of green chemistry. The described process is achieved and highly handled for all parameters of fabrication such as maximum yield and elevated purity growth control. It is the easiest process ever known which claims to produce high-quality CNTs by saving time, money, energy, and the environment [128].

7.9. Applications of Smart Materials Based on Carbon Nanotubes and Nanofibers

Industrial development requires a constant need for advanced materials for the advancement of science and technology. For fully filing with the ever-changing and increasing demand of better and advanced materials, the production of hybrid materials is a superior option.

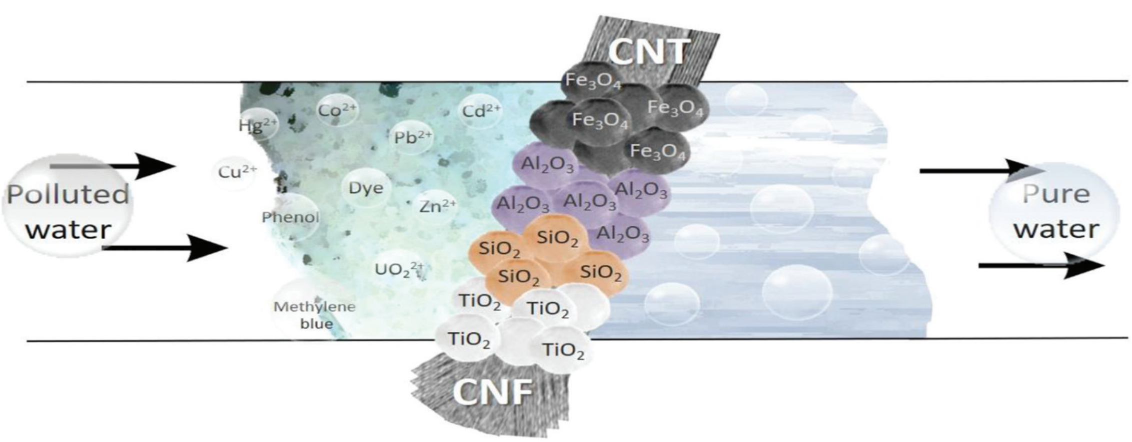

Figure 12. The figure depicts the application of hybrid materials in water purification. Hybrid materials are the synergy of two or more individual materials for the production of advanced materials (Reprinted with permission from Ref [129]. **Copyright 2020 Frontiers in Chemistry).**

Hybrid materials (HMs) are formed by combinations of different structural components which inherit some of their properties and functions [129]. After a combination of two or more different materials, the new material is structurally different from that of their constituent materials. Among many hybrid materials in the category of filamentous forms, hydride materials formed with carbon nanotubes and carbon nanofibers and some inorganic nanoparticles create a center of attention. Its structure and morphology, unique charge and energy transfer processes lead to synergistic effects. The review of these types of materials is useful in various ecological applications [130].

Figure 12 depicts the picture of the hybrid materials fabricated and utilized for environmental applications. The carbon nanofibers (CNFs), and carbon nanotubes (CNTs) and inorganic nanoparticles form a filamentous structure which render to a high specific surface area, unique properties better than the conventional nanomaterials.

Hybrid materials are the synergy of two or more individual materials for the production of advanced materials. It provides an additional degree of freedom which opens new avenues for the development of new materials. Hybrid materials have enhanced properties such as good conductivity, high sorption, better catalytic, mechanical, optical, and magnetic properties [131]. The hybrid materials manufactured by carbon nanotubes and carbon nanofibers in blend with inorganic (metal oxide) nanoparticles can be able to solve the maximum number of water treatment problems, helpful in preventing air pollution, and recycling of waste. These hydride materials act as proficient sorbents and help in photo catalysis which reduces the usage of materials. Environment electronic magnetic applications of hybrid materials could only be possible by developing enhanced compatibility among the materials used in making hybrids [132]. There is a need of development of novel protocols which proliferate new ideas for an inexpensive and trustworthy approach to the fabrication of advanced hybrid materials [133].

7.10. Diverse Applications, Present Trends, and Future Viewpoints of Nanotechnology in Food Science

Nanotechnology is a face-changing technology in various applications of food, medicine, and agriculture sectors [134]. Qualitative and quantitative production of healthier, safer food can be easily

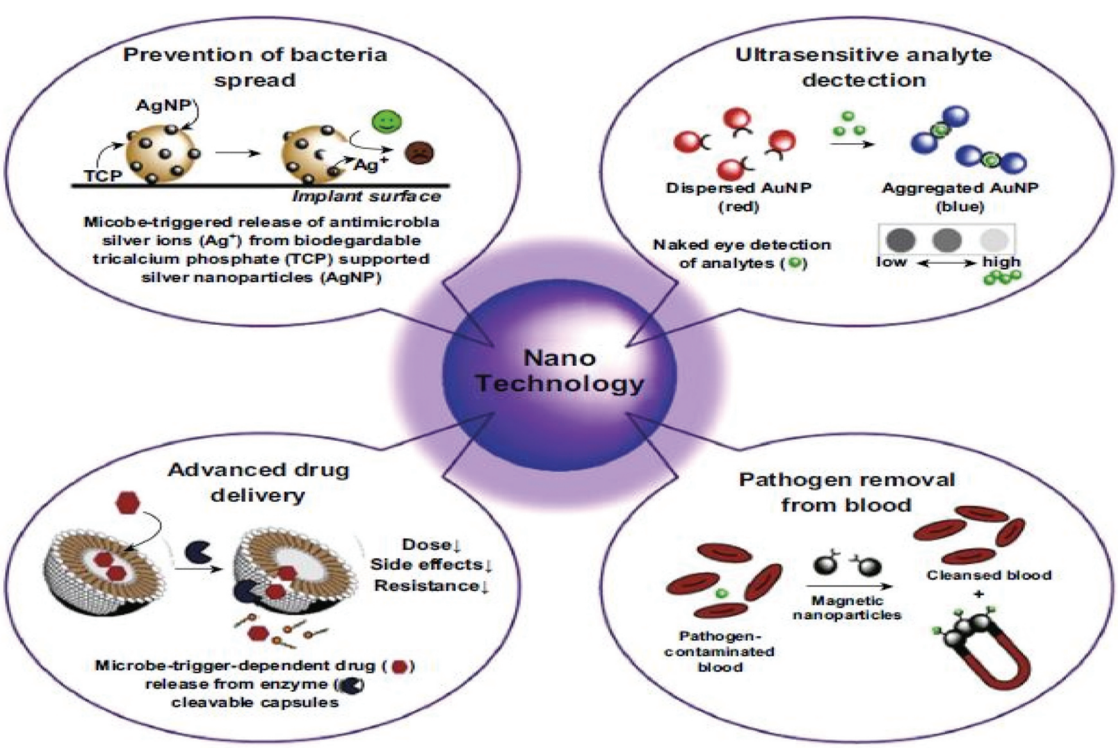

Figure 13. Nanotechnology-based solutions for the detection and prevention of microbial infections. In the figure, prevention of bacteria spread, ultrasensitiveanalyte advanced drug delivery, and pathogen removal from blood has been depicted (Reprinted with permission from Ref [134]. Copyright 2020 Springer Nature).

produced with the help of nanotechnology. Nanotechnology is helpful in food technology in various ways such as it increases the shelf life of food, checks, contamination, and produce of superior food quality. Better-quality food can be produced by nanotechnologies that are unpreserved or semipreserved in nature [135]. Conventional food technologies have certain voids which are easily filled by these nanotechnologies. Nanotechnologies are helpful in food technology in various arenas such as processing, packaging, security, and storage. It also helps in enhancing the esthetic look of the food. It increases the taste and consistency of food through inner modifications of particle size, surface charge, etc. Nanotechnology is also useful in food protection and conservation through advanced methods of food packaging, etc [136].

Nanotechnology is helpful in preventing, detecting, and treating bacterial infections. Various metallic nanoparticles are silver, gold are helpful in prevention, ultrasensitive detection, drug delivery, and pathogen removal from blood. Figure 13 clearly depicts the nanotechnology solutions usage at every stage in preventing disease drug delivery and detection.

Right from the food production, processing, storage, conversation, and packaging of food, nanotechnology plays a pivotal role [137]. Nanosensors help in detecting the stability of food, nutritional status, etc. Generally, food is biodegradable in nature, nanotechnologies raises the shelf life of food. Targeted deliverance of food bioactive compounds can be easily achieved by nanotechnology. Nanotechnology provides nano foods which are widely accepted by consumers. Uses of nanocolloidal particles in diverse branches of the food industry are prevalent. Sustainability and enhanced quality of foods can be achieved by colloidal particles in nanotechnology [138]. In a few countries, standard regulatory rules are implemented for the maximum use of nanotechnology in food products. Insufficient scientific investigations generate ambiguity among the producers as well as consumers and about the efficacy of nanosystems. Advancement of technologies like nanotechnology also creates a danger that it may come into the food chain through various sources of air, water, and earth. It can also damage DNA as well as cell death, etc. In vivo studies of nano foods in humans and animals may help in leading to certain conclusion. Scientists suggested that if nanotechnology is used in a regulated manner, it has been supporting and enhancing the quality of food which will benefit human health in the long run.

7.11. Conversation of Low Value Biomass with High Activated Carbon and Carbon Nanotubes

In the case study, carbon nanotubes were synthesized from low-value Miscanthus giganteus lignocellulosic biomass. The CNTs are synthesized from high-value materials out of low-value products. The characterization techniques such as XRD, SEM, FTIR, thermal analysis, TGA/DSC, and FTIR techniques were given here fully explained and provide testimony of the conversion of Miscanthus×giganteus lignocellulosic biomass into CNTs.

Universal energy requirement is accomplished by the constant use of fossil fuels. However, the use of fossil fuels replenishes our renewable resources at a very high speed. Certainly, there is a need for alternative fuels with low CO_2 and particulates to save our renewable resources and environment. The low biomass Miscanthus giganteus lignocellulosic has been converted into high-value materials such as activated carbon and carbon nanotubes. The

Miscanthus contains high alkali silicate content which generates a high amount of silica ash, emits NOx, SOx, and particulates [139]. The emissions related are spoiling our health, energy resources, and environment. An alternative approach has been tried here in which activated carbon has been synthesized by low biomass (Miscanthus giganteus). The formed activated carbon enhanced physical and chemical properties such as high surface area and pore volume of around $0.93\ cm^3 g{-}1$.

Utilization of waste biomass into useful products is an illustration of a circular economy. Circular economy means using waste product in generating useful products which in turn reduces pressure on the environment, safe renewable energy, utilization of waste, hence constant supply of raw materials, motivating innovation, making better economic growth. Waste lignocellulosic biomass has been converted into high surface area activated carbon multiwalled carbon nanotubes. To get the desired result, activated carbon with a contact angle of around $\theta = 9.89°$ was mixed with nitrogen-based material and iron precursor. The synthesized materials were tested for heavy metal removal tests using a lead solution, *90%* removal capacity has been seen after the first hour of testing. The material form may have a wide range of applications in the arena of wastewater treatment and other uses where activated carbon/carbon nanotubes are frequently used [140].

Table 1. Potential applications of CNTs in various fields.

Sr. No	Name of Material	Application in specific field	Roles	Reference
1	**Carbon nanotubes for water purification**	**Water purification**	Carbon nanotubes are able to remove all types of pollutants in organic, inorganic, or biological materials	[45]
2	**Carbon nanotubes in electrical/ electronic applications**	**Electrical/electronic applications**	Carbon nanotubes provide small size, carbon based nano-electronics having high flexibility compared to conservative silicon electronics. Carbon nanotubes have unusual properties because of the low dimensionality	[61]
3	**Carbon nanotubes in electrode materials**	**Electrode materials**	The CNT is also used as an electrode material for the capacitor to augment the ability of the capacitor to store higher energy density because of the large surface area	[63]
4	**Carbon nanotubes in energy conversion storage applications**	**Energy conversion storage applications**	The CNTs material has commenced ground-breaking developments in the field of energy conversion, which includes the development of solar cells, fuel cells, hydrogen storage, lithium ion batteries, and electrochemical super the capacitors.	[67]
5	**Carbon nanotubes as electrochemical and electronic biosensors**	**Electrochemical and electronic biosensors**	The CNTs improve electron transfer which is very useful in combining electrochemical and electronic biosensors	[70,71]
6	**CNT is used for targeted therapies such as drugs, gene delivery, and cancer treatment, other biomedical applications**	**Drugs, gene delivery and cancer treatment, other biomedical applications**	CNTs are effectively used in drug delivery systems and act as nano carriers. Versatile properties of carbon nanotubes make it suitable for biomedical applications. They are used in delivering drugs for anticancer drugs, genes, and proteins	[76]
7	**Carbon nanotube filters used in controlling air pollutants**	**Air pollutants**	Carbon nanotubes (CNTs) are novel materials used for controlling pollutant emissions because of the unique chemical and physical properties. Carbon nanotubes (CNTs) are used as sorbents for controlling environment pollutions	[93–96]
8	**Carbon nanotubes increase agricultural yields**	**Agricultural sector**	CNTs play a supporting role in both in vitro and in vivo. Both types of CNTs such as MWCNTs and SWCNTs are useful in delivering nutrients to plants as they are able to cross the plant cell walls and reach cellular organelles.	[97–99]
9	**Carbon nanotubes in reducing environmental pollution**	**Reducing environmental pollution**	CNTs are painstaking eco-friendly materials which reduce carbon footprint in the environment	[4,5]
10	**Environmental applications of carbon nanotubes sponge**	**CNTs sponge in the cleanup of oil from water**	The CNTs sponge is useful in removing oil in large quantities, i.e., *80* to *180* times their own weight, and can effectively be used in an extensive range of solvents and oils.	[110]
11	**Hybrid SWNTs ceramic filters has been developed and open new avenues in this field.**	**They are having tapered diameter, large surface area, and good thermal resistance which combines the porous aggregate structure of SWNTs.**	Efforts are made to earn people approval and enhance the arena nanomaterials, which is helpful in creating conducive environment	[112–184]
12	**Green carbon nanotubes**	**Carbon nanotubes prepared using organic camphor**	Elevated-purity carbon nanotubes (CNTs) can be produced from camphor hydrocarbons using chemical vapor deposition method	[113–115]
13	**Smart materials based on carbon nanotubes and nanofibers**	**Hybrid materials**	Hybrid materials (HMs) are formed by combinations of different structural components which inherit some of their properties and functions	[118]
14	**Nanotechnology in food science**	**Food Science**	Nanotechnology is also useful in food protection and conservation through advanced methods of food packaging, etc.	[125]
15	**Waste biomass into useful products with high activated carbon and carbon nanotubes**	**waste biomass into useful products with high activated carbon and carbon nanotubes**	Such CNTs have wide applications in the field of water adsorption applications, wastewater remediation, numerous industrial applications, for example, microelectronics, tissue engineering, composites, etc.	[128–132]

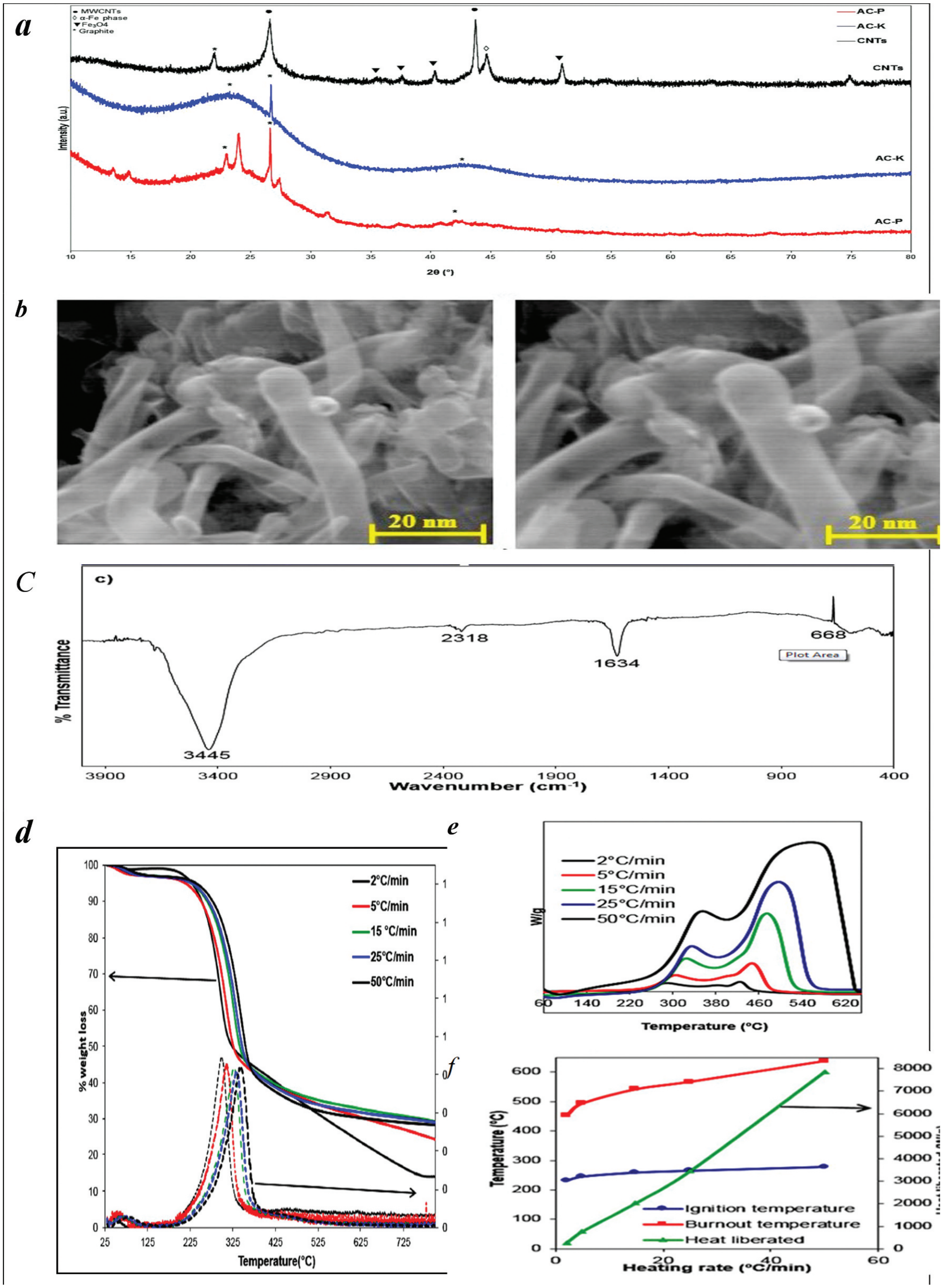

Figure 14. (a). The XRD diffractogram patterns of carbon nanotubes. (b). The SEM images of carbon nanotubes. (c) shows the FT-IR of the CNTs in the range of wave numbers of 400–4000 cm-1. (d) shows the thermal analysis under the N_2 atmosphere (e). The differential scanning calorimetry (DSC) curves with different heating rates under the N_2 atmosphere. (f) DSC curves with different heating rates under air atmosphere along with (g) the calculated ignition, burnout temperatures and heat liberated during the combustion (Reprinted with permission from Ref [139]. Copyright 2020 Springer Nature).

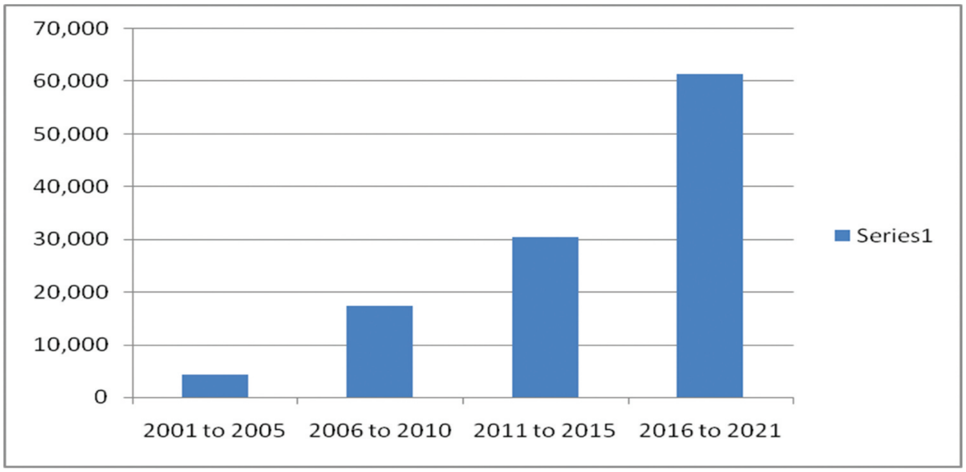

Figure 15. The trend of patents filed in the years [2001–2020].

7. Future Applications of CNTs

In this review article, the recent applications of CNTs in various fields, such as in water purification, water disinfection, electrical and electronic applications, biomedical energy, air purification, etc. have been discussed. They have given the 3 R (reduce, reuse, and recycle) concept. At the end, it is considered that the CNTs have been in use for decades for various applications. Nanomaterials, especially CNTs, act as an alternative material for future sustainable energy sources. The CNTs used in various applications serve as an excellent adsorbent, energy storage material, and energy transfer material. It can also be exploited in various applications of biomedical fields, such as biosensor, drug delivery, cancer treatment, etc. The CNTs are already commercially available and it is inevitable to escape from the application of CNTs. These technologies such as CNTs will hopefully open the gate to more successful and less costly toxicant technologies. As it can be seen, although, the success of these CNTs is reliant on a better understanding of their potential health impacts. The future application of CNTs have been discussed in Figure 15, which briefly portraits the trend of patents filed from 2001 to 2020. Summary of potential applications of CNTs in various fields are illustrated in Table 1 given below.

8. Conclusions

The material used in the future requires an extraordinary quality such as extraordinary mechanical, thermal, and chemical properties which can resist the adverse conditions of the environment as well as economical in use. The carbon nanotubes (CNTs) reinforced functionally graded composite material (FGCM) is a new generation material with uncharted potential and can be effectively used in various applications, such as aerospace, biomedical, industrial, and chemical applications. FGCM have numerous applications as its reinforced materials with the use of nanotechnology. Such types of materials are in demand and CNTs serve as an excellent candidate for future use [141,142].

To analysis the use of materials in the present and future, patent documents serve as a testimony of the materials. It helps to check the importance of the material in its current and future trends. It also serves as a source of data for use in innovation, technological change, and research and development. In the concluding paragraph, Figure 14 shows the trend of patents filed from 2001 to 2020. The data was collected from Derwent World Patents Index. The analysis not only proves that CNTs is an important material in the field of research and development but also provides a comprehensive picture of the CNTs innovation in the last 20 years [143–147].

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Disclosure Of Potential Conflicts Of Interest

No potential conflict of interest was reported by the author(s).

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